

DEEP LEARNING PRIMER - SAMPLE DOCUMENT (educational, for testing)

What is Deep Learning?

Deep learning is a subset of machine learning. It uses neural networks with many layers to learn patterns from data. Training optimizes a loss function using gradient descent. Deep learning powers image recognition, speech, and language applications.

Neural Networks

A neural network consists of layers of neurons. Each neuron has weights and a bias. Activation functions such as ReLU and sigmoid introduce nonlinearity. Common architectures include dense layers, convolutional layers, and recurrent layers.

Training

Networks are trained with backpropagation. Data is processed in batches over many epochs. The learning rate controls step size during gradient descent. Overfitting occurs when the model memorizes the training data.

Regularization

Regularization methods reduce overfitting. Dropout randomly drops neurons during training. L2 regularization penalizes large weights. Early stopping ends training when validation performance stops improving.

Convolutional Networks

Convolutional neural networks (CNNs) are designed for images. They use kernels to detect features and pooling layers to reduce spatial size. CNNs achieve strong results on object recognition tasks.

Transformers

Transformers rely on the attention mechanism. Attention weighs how much each token relates to every other token. Transformer models form the basis of modern large language models. They process sequences in parallel, which accelerates training.